

The Impact of Remote Work on Software Team Productivity

Abstract

This paper examines productivity outcomes across 40 distributed software engineering teams over an 18-month period. We find that teams with structured asynchronous communication protocols showed a 23% increase in shipped features compared to teams relying primarily on synchronous meetings.

Introduction

Remote work adoption accelerated dramatically starting in 2020 and has remained a dominant mode of operation for software organizations since. Prior research has focused primarily on individual productivity metrics, leaving a gap in understanding team-level dynamics. This study addresses that gap by tracking cross-functional engineering teams across three company sizes: startup (under 50 employees), mid-market (50-500 employees), and enterprise (500+ employees).

Methodology

We collected data from 40 teams using a combination of git commit analysis, sprint velocity tracking, and quarterly engineer surveys. Teams were categorized by their dominant communication style: synchronous-first (daily standups, frequent video calls) versus asynchronous-first (written updates, threaded discussions, core-hours overlap of 2-3 hours only). Data was normalized for team size and seniority mix.

Key Findings

Asynchronous-first teams shipped 23% more features per quarter on average, with the effect most pronounced in teams spanning more than 3 time zones. However, asynchronous-first teams also reported 15% lower scores on 'sense of team cohesion' in quarterly surveys, suggesting a tradeoff between raw output and interpersonal connection. Enterprise-size teams showed smaller productivity gains (11%) compared to startups (31%), likely due to existing process overhead in larger organizations.

Conclusion

Organizations optimizing purely for throughput should consider structured asynchronous workflows, particularly for globally distributed teams. However, leaders should pair this shift with deliberate investment in team cohesion activities to offset the connection deficit observed in our survey data.